

Three ways an LLM looked better than it was

Removing one confound at a time from an LLM-guided quantum circuit search

Headline: each apparent advantage came from something other than design skill - circuit size, then a bad starting point. The LLM beats a textbook ansatz on all four tasks and never beats random search on any (30 seeds).

Tomoya Hatanaka

ML4SCI / Google Summer of Code 2026

July 31, 2026 · commit 3f36d8c · experiment/bench-v2-real-llm-cost-minimal

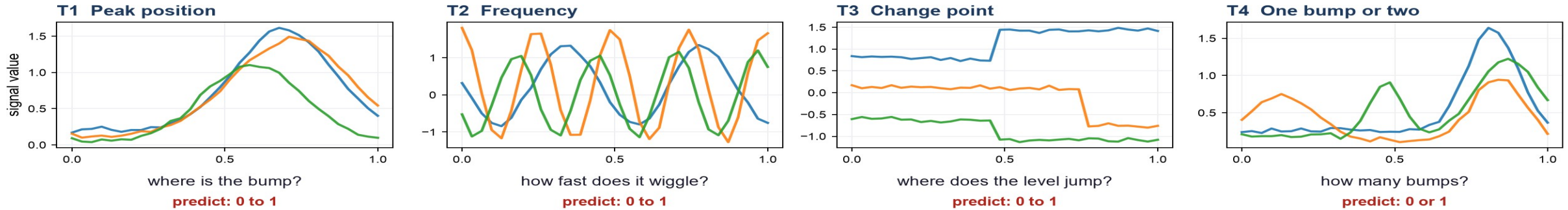
WHAT CHANGED

Last week the LLM could choose how big the circuit was



Longer circuits usually score better. A proposer that always asks for the maximum length wins without designing anything. Fixing the length removes that move; 16 gates then makes the circuit visible to the readout.

Four questions about a 32-number signal



Each signal is exactly 32 numbers - the amount 5 qubits hold. Three examples per panel; line colour only separates the examples.

All four are synthetic

so difficulty and leakage are controlled,
and the answer is known exactly

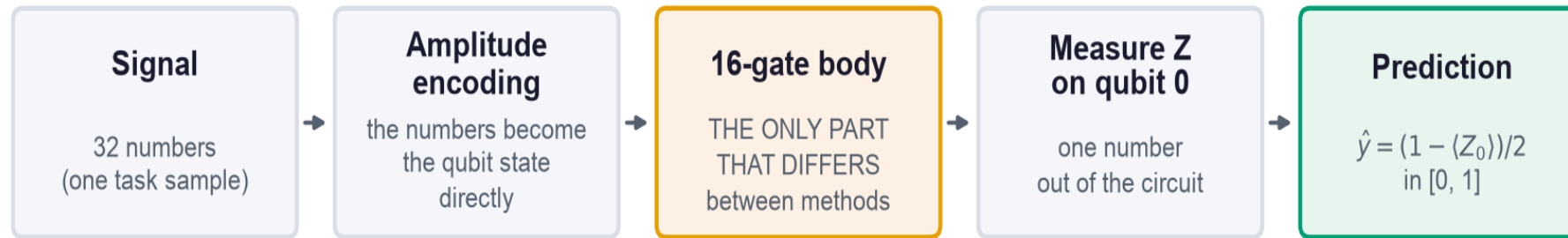
T1-T3 are regression

the target is a position or a rate, so
RMSE is the natural score

T4 is a yes/no question

RMSE there is the error against a 0/1
label; AUROC is recorded too

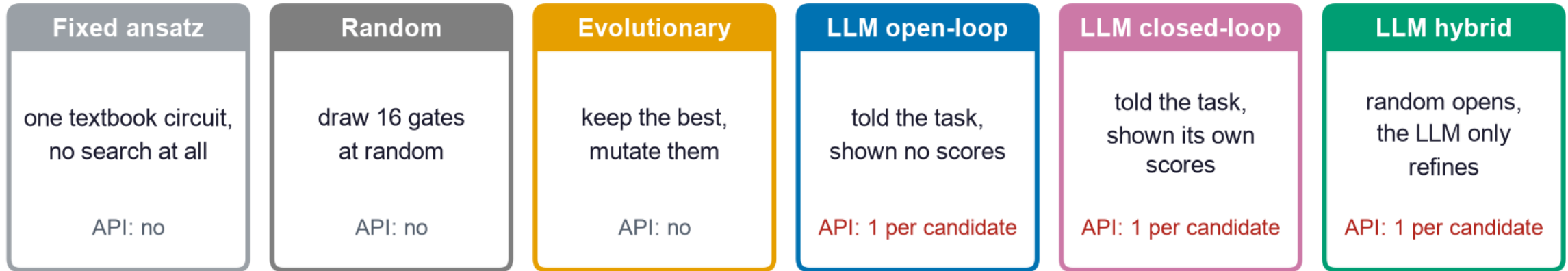
One pipeline, and every constraint on it



No training: a method picks the **RANGE** each angle comes from and the harness draws inside it. **No classical layers:** nothing outside the circuit can fix a bad circuit.

Constraint	Value
Gate count	16 exactly - enough that the measured qubit can see the circuit (slide 7)
Gate set	H, RX, RY, RZ (one qubit) · CRX, CRY, CRZ (two qubits)
Angles	the method picks one of 8 RANGES over $[-\pi, \pi]$; the harness draws inside it
Budget B	8 distinct circuits scored per method · 30 seeds
Scoring	validation RMSE picks the winner; test RMSE is read once, after that choice

Six methods, one shared budget



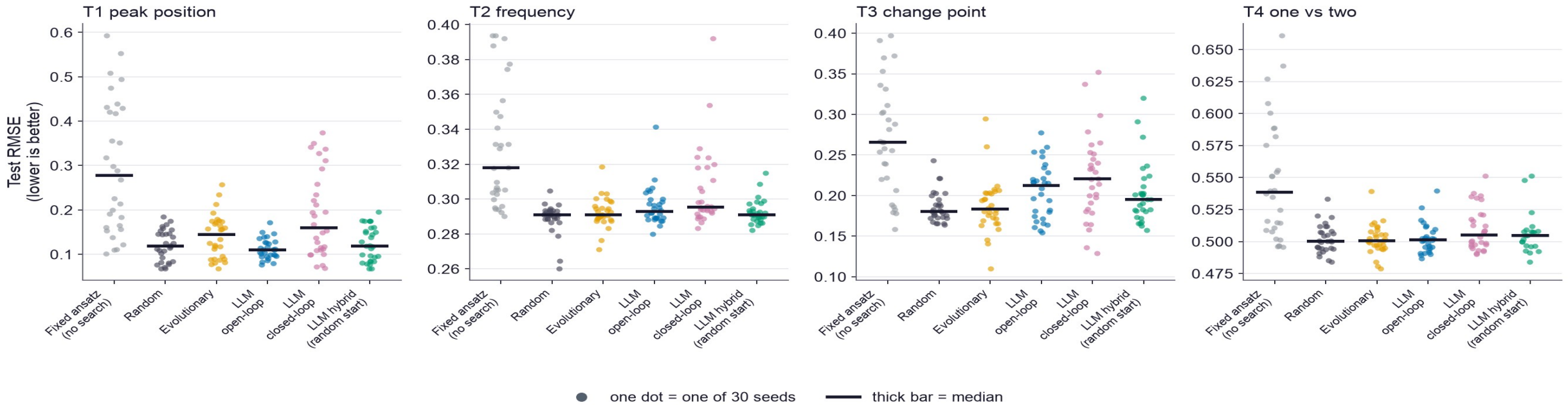
All six get the same budget $B = 8$: the same number of DISTINCT circuits actually scored. The fixed ansatz searches nothing, so it spends one.

All three LLM arms are told what the task is. Open-loop never sees how its circuits scored, closed-loop does, and the hybrid lets random pick the first two before the LLM takes over.

RESULT

16 gates, 5 qubits, $B = 8$, 30 seeds all four tasks

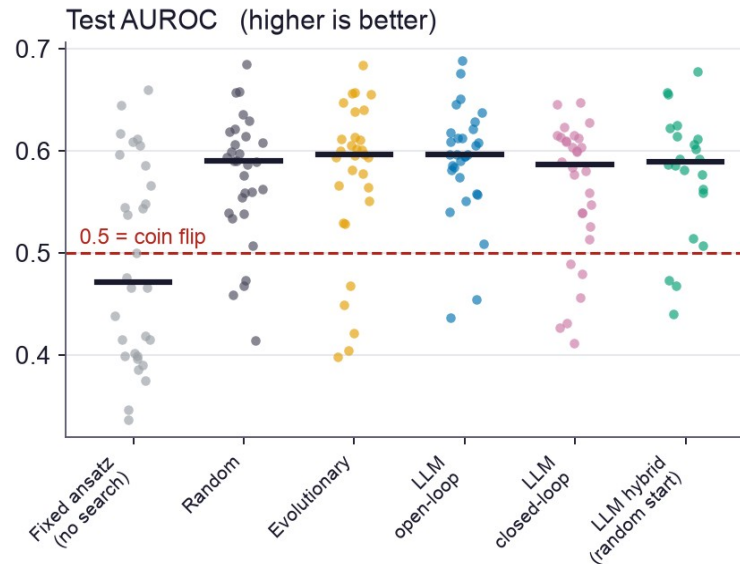
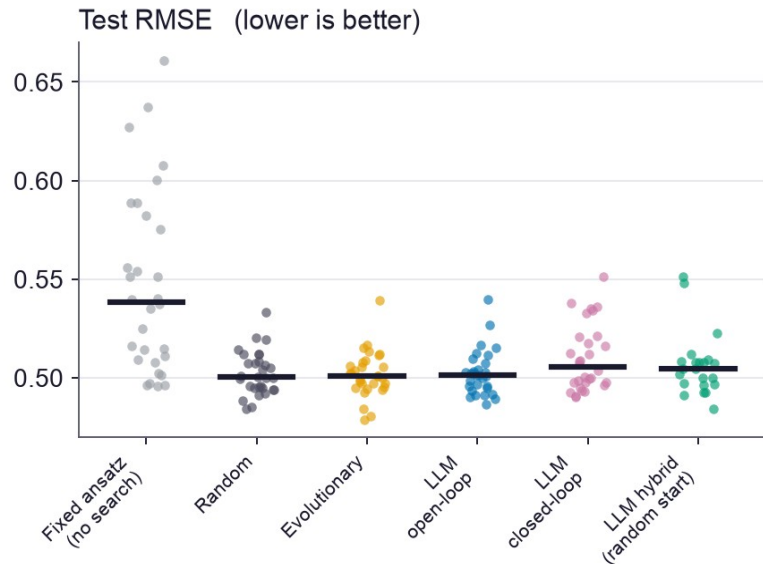
The LLM beats the textbook ansatz, never beats random



Every panel has its own y-axis. RMSE compares arms inside a panel, never across panels.

vs the fixed ansatz: significant on all 4 tasks. **vs random: no difference detected on any.** Random holds the better median on T2, T3 and T4 (T3 0.180 vs 0.212); the LLM leads only on T1 (0.110 vs 0.119), and not significantly.

On T4 the two scores tell different stories



AUROC

= how often the model ranks a two-bump signal above a one-bump one. 1.0 perfect, 0.5 a coin flip.

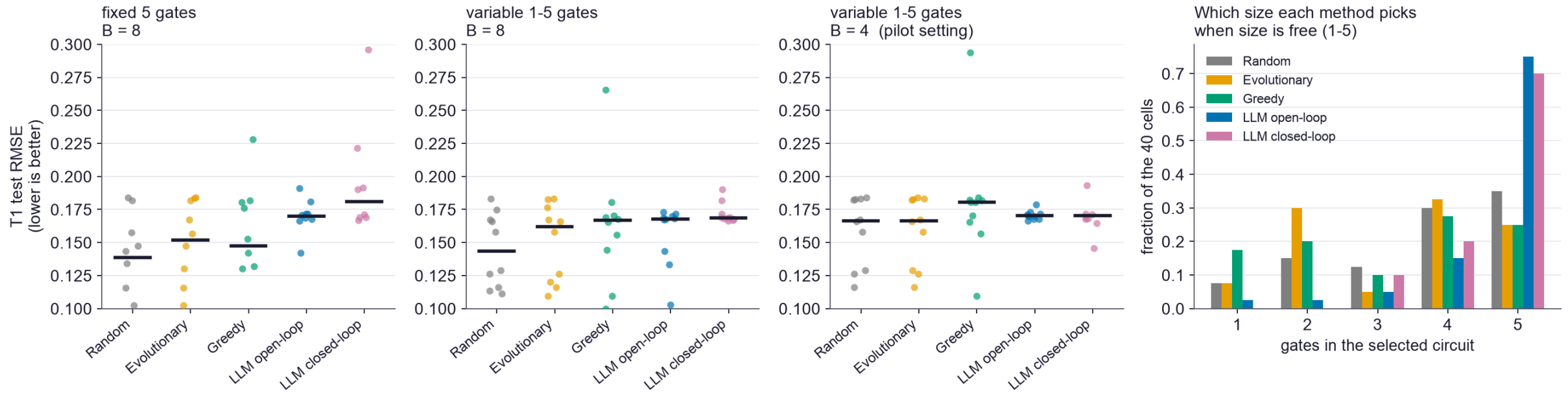
Why report both

RMSE also punishes outputting 0.45 / 0.55 instead of 0 / 1. AUROC only cares about the order, so one alone can assert the wrong winner.

RMSE against a 0/1 label is a proper score, but it also punishes a circuit that ranks the classes correctly while outputting 0.45 and 0.55. AUROC only cares about the order, so both are shown.

Neither reading is strong. Every method sits near 0.50 RMSE and barely above chance on AUROC. Sixteen gates without training do not solve T4, and no arm separates from the others.

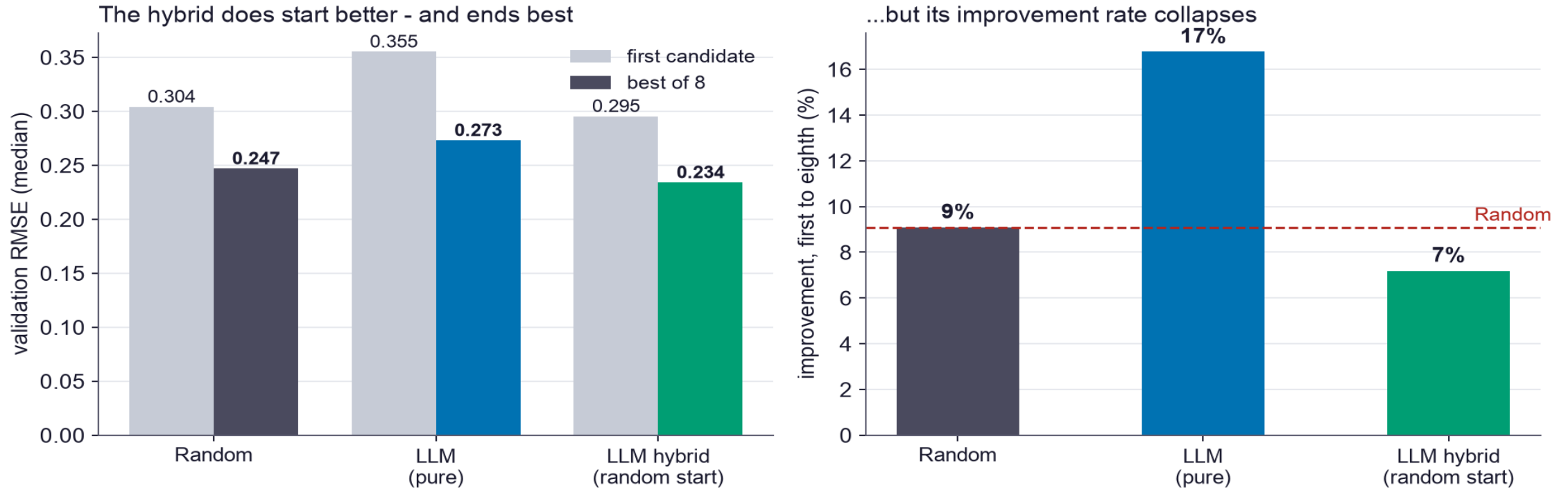
Last week's advantage was a size decision, not a design one



Left three panels: one dot = one of 10 seeds, thick bar = median. Only Random degrades as circuit size becomes a free choice, because the LLM almost always picks the maximum size while Random draws it uniformly. The pilot rewarded that single decision.

Free the length and only Random gets worse (0.139 → 0.167); the LLM does not move (0.170 → 0.170), because it was already asking for 5. At the pilot's setting they meet.

The LLM's "refinement skill" was the room a bad start leaves



The pure LLM improved 17% where random improved 9%, which looked like refinement skill. Give it a good start and that rate falls to 7%, below random. The 17% was the room a bad start leaves, not an ability.

I built the hybrid arm expecting it to win. It reached the best final score of any arm and still could not beat random - because the ability it was built to exploit was not there.

What I can and cannot claim

Supported

- The LLM beats a textbook ansatz on all 4 tasks
- It never beats random search - 30 seeds, 12 pre-declared contrasts
- Given free length it picks the maximum ~70% of the time
- Its apparent refinement skill vanishes given a good start

NOT supported

- "LLMs cannot design circuits" - one small model, $B = 8$
- "Task context does not help" - never A/B tested at equal seeds
- Anything about hardware - this is exact simulation
- Comparing RMSE across different tasks

Next: give the LLM a decision that is actually hard, and check any advantage the same way.